Adding Two n-Bit Integers

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Adding Two n-Bit Integers:

Suppose A and B are two n-bit integers:

$$A = (A_{n-1}A_{n-2}\dots A_0)$$

$$B = (B_{n-1}B_{n-2}\dots B_0)$$

where $A_i, B_i \in \{0, 1\}$. To add A and B, we use n Full Adders connected in series (Ripple Carry Adder).

1 Operation of a Full Adder

A **Full Adder** is a combinational circuit that computes the arithmetic sum of three binary bits (two input bits x, y and an input carry c_{in}). It produces two outputs: a **Sum (S)** bit and a **Carry-out (c_{out})** bit.

Truth Table for Full Adder:

x	y	c_{in}	S	c_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Output:

i) Sum (S)

$$\begin{aligned} S &= x'y'c_{\text{in}} + x'yc_{\text{in}}' + xy'c_{\text{in}}' + xyc_{\text{in}} \\ &= x'(y \oplus c_{\text{in}}) + x(y \oplus c_{\text{in}})' \\ &= x \oplus y \oplus c_{\text{in}} \end{aligned}$$

ii) Carry-out (c_{out})

$$\begin{aligned} c_{\text{out}} &= x'yc_{\text{in}} + xy'c_{\text{in}} + xyc_{\text{in}}' + xyc_{\text{in}} \\ &= xy + c_{\text{in}}(x \oplus y) \end{aligned}$$

2 Binary adder

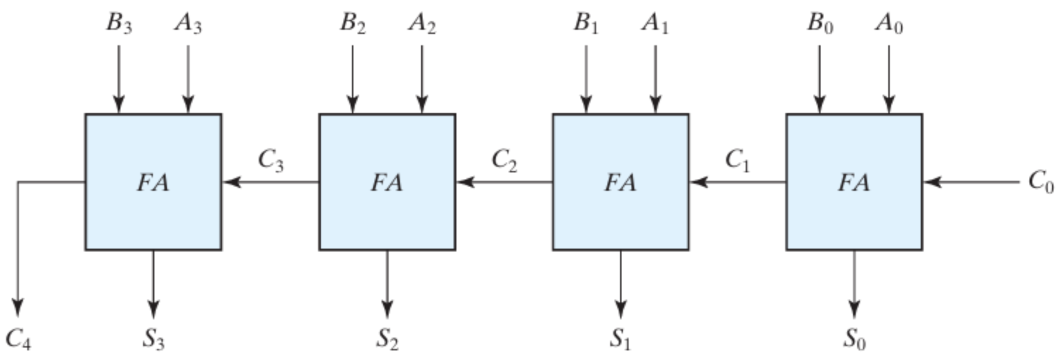
For the i-th bit position ($0 \le i \le n-1$), the Full Adder receives input bits A_i, B_i and carry-in C_i, and computes:

- **Sum bit:** $S_i = A_i \oplus B_i \oplus C_i$
- **Carry output:** $C_{i+1} = A_iB_i + C_i(A_i \oplus B_i)$

where $C_0 = 0$ and $C_n =$ final carry. So after n Full Adder finish, the final output is:

$$A + B = (C_n, S_{n-1}, ..., S_1, S_0)$$

Circuit for $n = 4$ is given below:

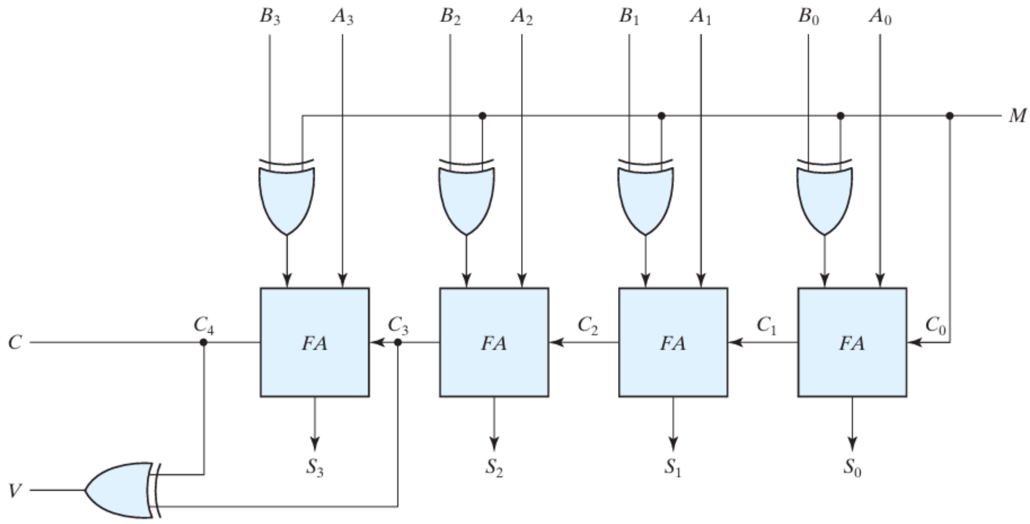


3 Binary Subtractor

The subtraction $A - B$ can be done by taking the 2's complement of B and adding it to A . The 2's complement can be obtained by taking the 1's complement and adding 1 to the least significant pair of bits. The 1's complement can be implemented with inverters, and a 1 can be added to the sum through the input carry.

So to perform $A - B$, inverters are placed between each data input B. The input carry C_0 must be equal to 1 when subtraction is performed. The operation becomes A plus the 2's complement of B.

NOTE: The addition and subtraction operations can be combined into one circuit with one common binary adder by including an XOR gate with each full adder. A four-bit adder–subtractor circuit is shown below:



Mode of operation: When $M = 0$, we have $B \oplus 0 = B$. So the FA_0 receives the correct value of b and input carry $C_0 = 0$ and the circuit performs $A + B$.
When $M = 1$, we have $B \oplus 1 = B'$ and input carry $C_0 = 1$. So all B inputs get complemented and 1 is added through the input carry and it performs $A - B$.

4 Complexity Analysis

- **Time Complexity:** $\mathcal{O}(n)$ — There are n Full adders. Each Full Addertakes a fixed amount of time to generate its carry-out. Since stage $i + 1$ can't computes its final carry until stage i finishes, the total time required grows linearly with n.

References

[1] Self-authored Class Notes, *Computing Systems Course*, Semester 1.

[2] M. Morris Mano and Michael D. Ciletti, *Digital Design: With an Introduction to the Verilog HDL*, Pearson, 5th Edition, 2012.